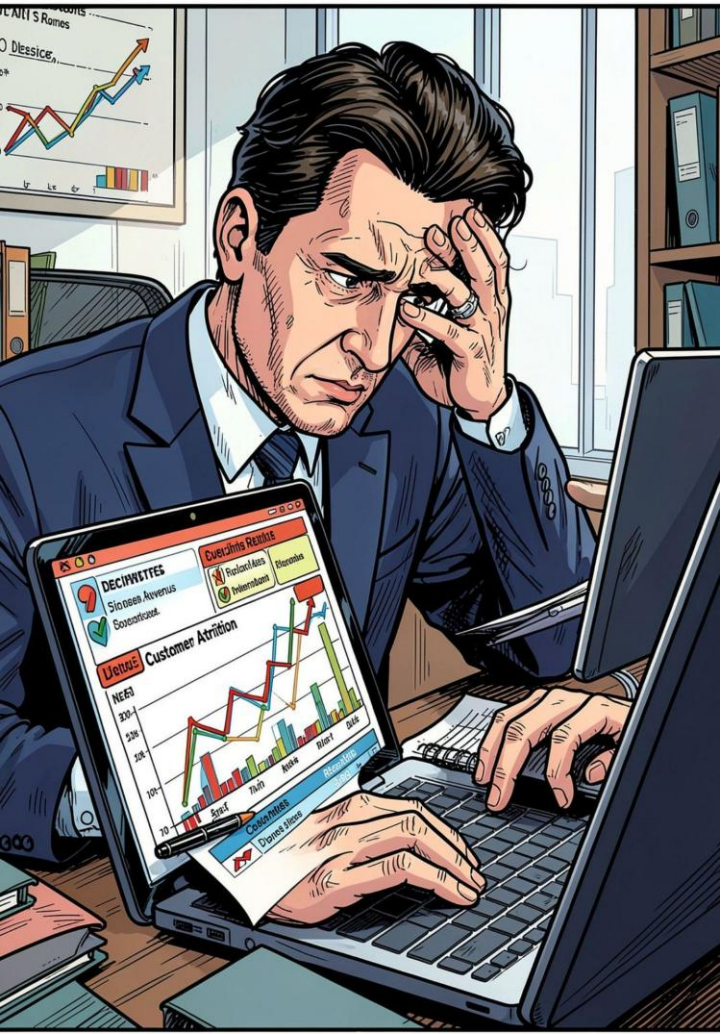
Customer Churn Prediction System

Subscription-Based Services · Behavioral Analytics · Machine Learning

ACADEMIC PROJECT PRESENTATION

BADIGANTI KANAKA MAHALAKSHMI
Reg No:-32312951004





The Retention Crisis in Subscription Business

Reactive Strategies

Traditional retention responds only after customers leave — too late to recover lost revenue.

Manual Analysis

Human-led behavior analysis cannot scale to millions of customers or detect subtle churn signals.

Delayed Detection

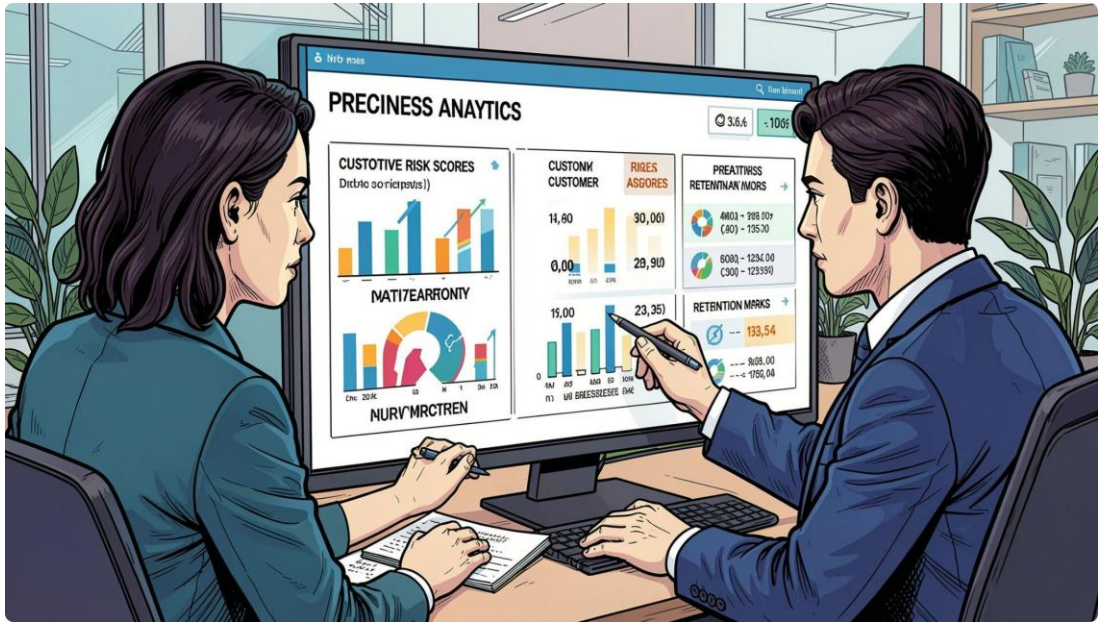
At-risk customers go unidentified until they cancel, eliminating any window for intervention.

Revenue Erosion

Customer attrition silently compounds, eroding recurring revenue and lifetime value over time.

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Why Accurate Churn Prediction Is Essential



Churn prediction transforms raw customer data into **actionable foresight** — enabling teams to intervene before attrition occurs.

Proactive Intervention

Identify at-risk customers weeks before they cancel.

Targeted Retention

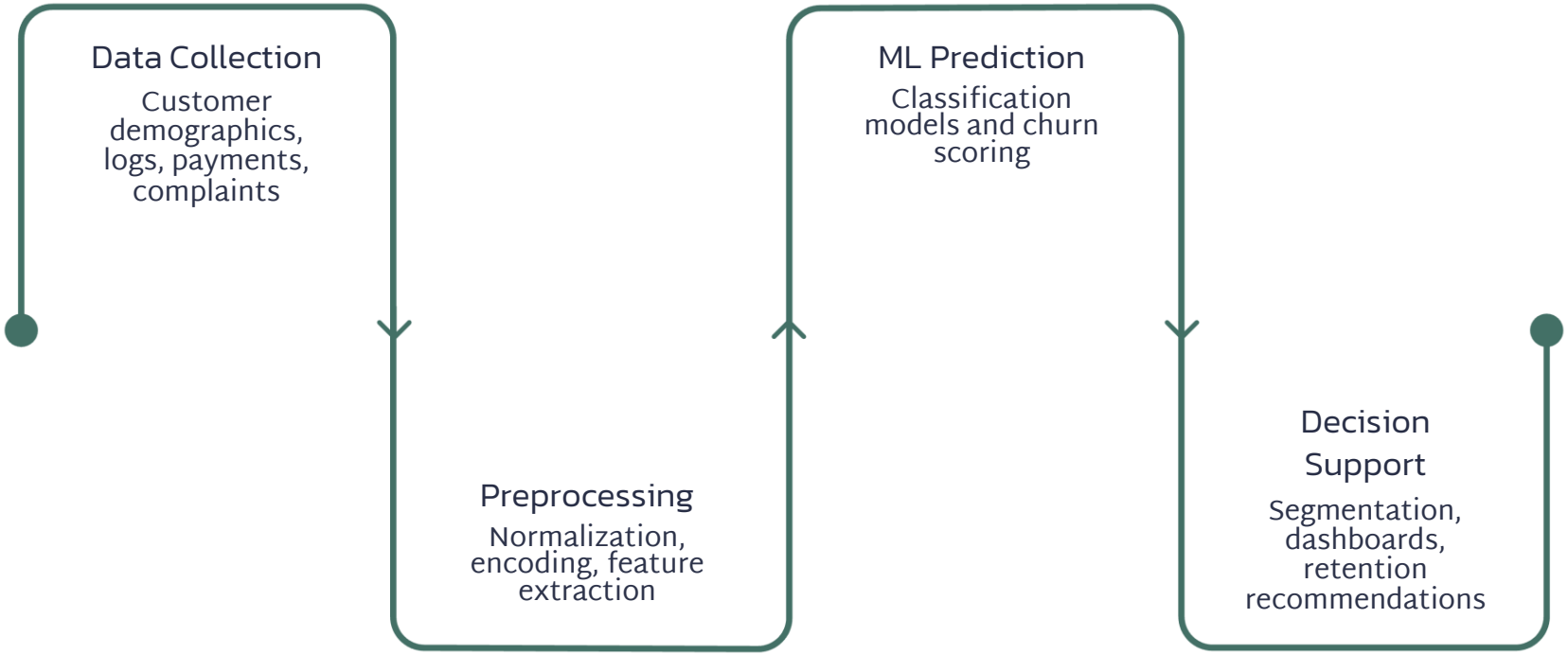
Deploy personalized offers to high-value, high-risk segments.

Revenue Protection

Reduce involuntary churn and protect recurring revenue streams.

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System Architecture Overview



The system integrates behavioral data from multiple touchpoints into a unified ML pipeline — delivering churn probability scores and retention recommendations through a secure web-based administration interface.

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Core Technologies & Concepts



ML Classification

Logistic regression, random forest, and gradient boosting classify customers as churners or retainers.



Behavioral Analytics

Usage frequency, session duration, and engagement patterns reveal early attrition signals.



Data Preprocessing

Missing value imputation, outlier handling, and normalization prepare raw data for modeling.



Feature Engineering

Derived features — recency, frequency, payment delinquency — amplify predictive signal.



Customer Segmentation

Clustering groups customers by behavior and risk profile for targeted retention.



Data Visualization

Interactive dashboards translate model outputs into business-ready insights.

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Key Platform Features

1

Customer Data Management

Centralized repository for demographics, subscription history, usage logs, and complaint records.

2

Predictive Model Training

Automated training, cross-validation, and performance evaluation with configurable thresholds.

3

Churn Probability Scoring

Every customer receives a real-time churn score, ranked and filterable by risk tier.

4

Retention Strategy Engine

AI-recommended interventions — discounts, outreach, or plan upgrades — matched to segment profiles.

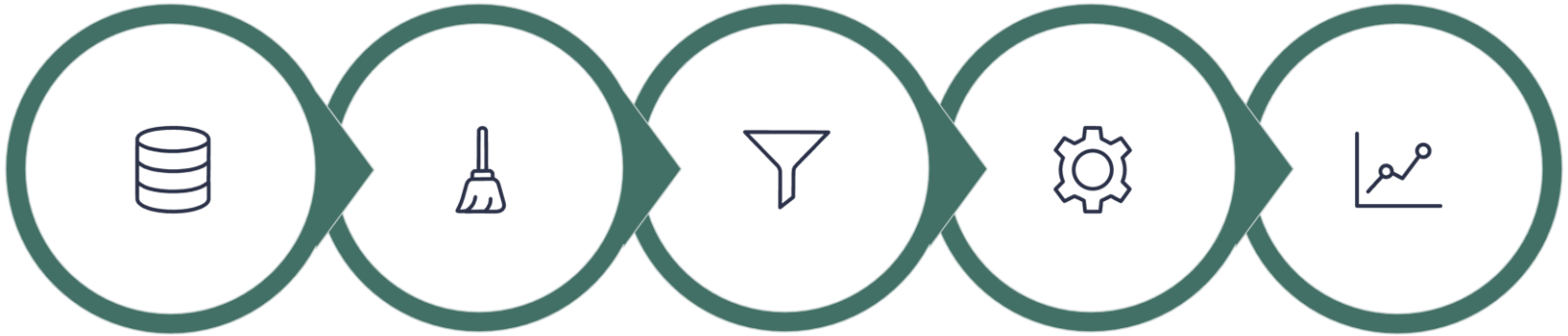
5

Analytical Dashboards

Reports on churn trends, model performance, campaign ROI, and customer behavior patterns.

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Implementation Methodology



Data Collection

Preprocessing

Feature Selection

Model Training

Prediction & Scoring

The end-to-end pipeline ensures data quality at each stage — from raw ingestion through model deployment — with rigorous evaluation metrics validating performance before production use.

Scalability & Security

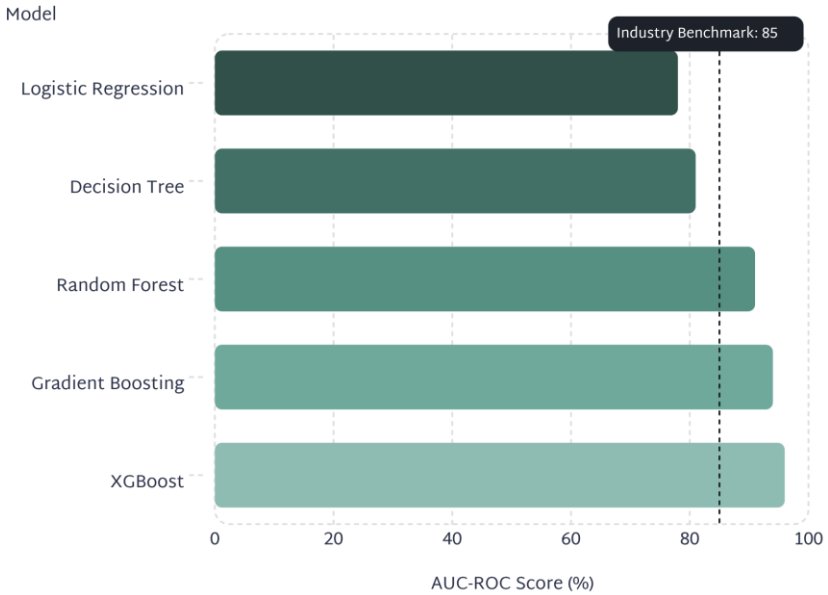
Designed to handle millions of records across industries with role-based access control and encrypted data storage.

Interpretability

Feature importance scores and SHAP values explain predictions, building trust with business stakeholders.

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Model Performance & Evaluation



Why Gradient Boosting Leads

Ensemble methods like XGBoost consistently outperform single-model approaches by combining multiple weak learners, reducing overfitting, and capturing complex non-linear patterns in customer behavior data.

Precision: 93%

Few false positives — retention resources focus on truly at-risk customers.

Recall: 89%

High capture rate of actual churners before they cancel.

F1-Score: 91%

Balanced precision and recall across diverse customer segments.

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Business Benefits Delivered

Improved Prediction Accuracy

ML models outperform rule-based heuristics in identifying at-risk customers.

Proactive Retention

Intervene before cancellation — not after — with timely, targeted outreach.

Reduced Revenue Loss

Protect recurring revenue by retaining high-value customers before they leave.

Optimized Campaigns

Focus marketing spend on segments with the highest retention ROI.

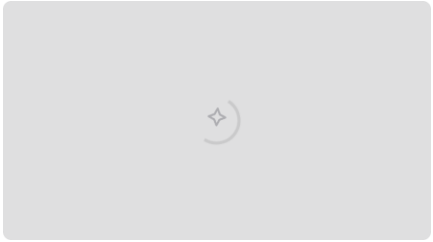
Increased Customer LTV

Longer retention cycles directly increase lifetime value and profitability.

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The Road Ahead: Future Enhancements

The system's architecture supports continuous evolution — from deeper models to real-time intelligence and enterprise integrations.



01

Deep Learning Models

Neural networks to capture complex temporal behavior patterns.

02

Real-Time Analytics

Streaming predictions triggered by live customer events.

03

Explainable AI

Transparent, interpretable predictions for stakeholder trust.

04

CRM & Marketing Integration

Native connectors for Salesforce, HubSpot, and automation platforms.

05

Cloud & Mobile Deployment

Scalable cloud infrastructure with mobile analytics access.

06

Sentiment Analysis

NLP-driven feedback analysis to enrich churn signals.

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